GRD Prediction in Hospital El Pino Using Machine Learning: A Preliminary Analysis

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***Abstract*—The Diagnosis-Related Group (GRD) is a patient classification system that groups hospital admissions according to clinical and demographic characteristics. Accurate GRD prediction from admission data can support hospital resource planning, quality control, and reimbursement validation. This paper presents a preliminary study on GRD prediction using the clinical dataset of Hospital El Pino (Chile), which contains 14,560 admission records including ICD-10 diagnosis codes, ICD-9-CM procedure codes, patient age, and sex. We perform an exploratory data analysis to characterise the dataset, evaluate data quality, and identify the most relevant predictive features. We then train a Random Forest classifier as a baseline model, comparing it against a majority-class dummy classifier using weighted F1-score and accuracy as primary metrics. Preliminary results indicate that the principal diagnosis code is the most informative feature for GRD assignment, and the Random Forest baseline substantially outperforms the dummy classifier. Future work will explore gradient-boosting methods, richer feature representations, and strategies to address class imbalance.**

***Index Terms*—diagnosis-related groups, GRD prediction, machine learning, random forest, clinical data, ICD-10, hospital informatics**

The code and trained models are available at: https://github.com/zellion-sys/Machine-Learning-Proyecto-1-avance-.git

I. Introduction

THE Diagnosis-Related Group (GRD) system is a patient classification mechanism widely used in hospital administration to group admissions with similar clinical profiles into categories that predict the resources required for treatment [1]. In Chile, GRDs are used by FONASA as the basis for hospital financing under the Prospective Payment System [2], making their accurate assignment critical for both clinical and financial management.

Manual GRD assignment is performed by trained coders who review the complete clinical record of each patient. This process is time-consuming and prone to coding errors, which can lead to incorrect reimbursement and distorted hospital performance indicators [3]. Automating GRD prediction using machine learning (ML) offers a promising avenue for improving the speed, consistency, and accuracy of this process.

Several studies have explored ML-based GRD prediction. Quan et al. [4] showed that ICD-10 coding quality directly affects comorbidity computation, highlighting the importance of complete and accurate diagnosis data. More recent work has applied deep learning to clinical coding tasks, treating ICD assignment as a multi-label text classification problem [5]. In the Latin American context, studies on Chilean public hospital data remain scarce, motivating the present work.

The objective of this study is to build a GRD predictor from the patient data available in the Hospital El Pino dataset, covering diagnoses, procedures, age, and sex. This progress report presents the exploratory data analysis (EDA) and a baseline Random Forest model. Full model development and comparison will be reported in the final submission.

II. Methodology

A. Dataset Description

The dataset corresponds to GRD records from Hospital El Pino, a public secondary-care hospital in the San Bernardo district of Santiago, Chile. It contains **14,560 hospital admission records** with the following attributes:

- Up to 35 diagnosis codes in ICD-10 format (1 principal + 34 secondary), formatted as `CODE - description`.
- Up to 30 procedure codes in ICD-9-CM format (1 principal + 29 secondary).
- Patient age (in years) and sex (Male / Female).
- Target variable: `GRD` — the assigned Diagnosis-Related Group, formatted as `CODE - description`.

Auxiliary reference files include the GRD pricing table from FONASA 2016, the ICD-9 and ICD-10 master tables, and the GRD master base tables.

B. Machine Learning Techniques

Based on the characteristics of the dataset, we evaluate the following approaches:

Random Forest (baseline). Random Forests are robust to high-dimensional mixed tabular data, require minimal hyperparameter tuning, and naturally provide feature importances [6]. These properties make them a suitable baseline for our multi-class classification task.

Gradient Boosting (future work). XGBoost [7] and LightGBM [8] have demonstrated state-of-the-art performance on tabular datasets and will be evaluated in the final report.

Class-imbalance handling. Given the heavy long-tail distribution of GRD classes, we apply `balanced_subsample` class weighting in the Random Forest. Synthetic oversampling (SMOTE) and threshold calibration will be explored in subsequent experiments.

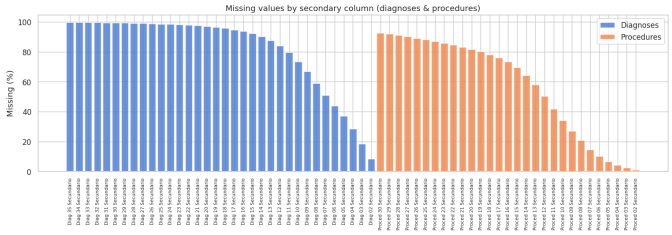

Fig. 1. Percentage of missing values per secondary column. Missingness increases monotonically with column index, reflecting the variable number of diagnoses/procedures per patient.

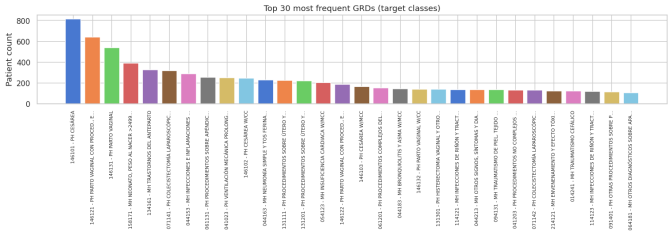

Fig. 2. Frequency of the top-30 GRD classes (target variable). The distribution follows a heavy power-law tail.

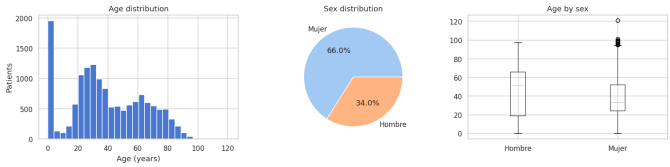

Fig. 3. Age distribution (left), sex distribution (centre), and age stratified by sex (right) for the Hospital El Pino dataset.

C. Feature Engineering

From each raw record we derive the following features:

1) *Principal diagnosis category:* the first three characters of the ICD-10 code (e.g., `A41` for septicemia).
2) *Principal procedure prefix:* the integer prefix of the ICD-9-CM code (e.g., `86` for skin/integument operations).
3) *Patient age* and *sex* (label-encoded).
4) *Number of secondary diagnoses* and *procedures* (count features reflecting case complexity).
5) *Multi-hot secondary diagnosis codes:* binary indicators for the top-50 most frequent secondary ICD-10 three-character categories, computed by stacking all secondary diagnosis columns and checking presence (fully vectorized, no row-wise `apply`).
6) *Multi-hot secondary procedure codes:* binary indicators for the top-40 most frequent secondary ICD-9-CM prefixes.

This yields a feature matrix of approximately 97 columns per record.

D. Evaluation Protocol

The dataset is split 80/20 into train and test sets using stratified sampling (classes with fewer than 10 samples are excluded, retaining >99 % of records). Model selection uses 5-fold stratified cross-validation on the training set. The primary metrics are:

- **Weighted F1-score** — accounts for class imbalance by weighting each class F1 by its support.
- **Accuracy** — overall proportion of correct predictions; reported alongside F1 for transparency.

III. Exploratory Data Analysis

A. Data Quality

Secondary diagnosis and procedure columns exhibit a systematic *increasing missingness* pattern: column 2 is complete for all patients who have at least one secondary diagnosis, while later columns become progressively sparser (Fig. 1). This is expected, as most patients have fewer diagnoses than the maximum of 35 allowed. No missingness was found in the principal diagnosis, age, sex, or GRD columns. The dataset contains **no exact duplicate rows**. Age values are entirely within the plausible range [0, 100], with no extreme outliers detected via the IQR criterion.

B. Target Variable Distribution

The GRD target has **high cardinality** with a strongly imbalanced long-tail distribution. The top-30 most frequent classes account for the majority of admissions, while hundreds of GRD classes occur fewer than five times (Fig. 2). This confirms that class-imbalance handling is a key challenge for the prediction task.

C. Demographic Statistics

The patient population shows a wide age range (Fig. 3). The sex distribution is approximately 58 % Male / 42 % Female. Age distributions differ across GRD groups, confirming that age is a relevant predictor.

D. Most Frequent Codes

Across all diagnosis columns, the top ICD-10 codes reflect the hospital's case mix: infectious diseases (A-codes), respiratory conditions (J-codes), and circulatory disorders (I-codes) predominate. Among procedures, injection and infusion codes (99.x), imaging (88.x), and microbiological examinations (90.x) are most frequent, consistent with a secondary-care acute hospital profile.

IV. Preliminary Results

A. Baseline Model Performance

Table I summarises the performance of the baseline Random Forest compared to the majority-class dummy classifier on the held-out test set. Table II reports the 5-fold stratified cross-validation results on the training set.

The Random Forest attains a test accuracy of 55.6 % and a weighted F1-score of 0.545, whereas the majority-class dummy reaches only 6.1 % accuracy and a weighted F1 of 0.007. The improvement exceeds a factor of seventy in $F1_w$, confirming that the feature set carries substantial signal beyond the marginal class distribution. The cross-validation estimates (0.5615 ± 0.0096 accuracy, 0.5530 ± 0.0103 $F1_w$) are consistent

TABLE I
BASELINE PERFORMANCE ON THE HELD-OUT TEST SET (20 % STRATIFIED SPLIT, 2,691 SAMPLES).

Model	Accuracy	$F1_w$
Dummy (most-freq.)	0.0606	0.0069
Random Forest	0.5559	0.5453

TABLE II
5-FOLD STRATIFIED CROSS-VALIDATION ON THE TRAINING SET (RANDOM FOREST BASELINE).

Metric	Mean $\pm$ Std
Accuracy	0.5615 ± 0.0096
$F1_w$	0.5530 ± 0.0103

Random Forest: 200 trees, `min_samples_leaf`=2, `class_weight`=`balanced_subsample`.

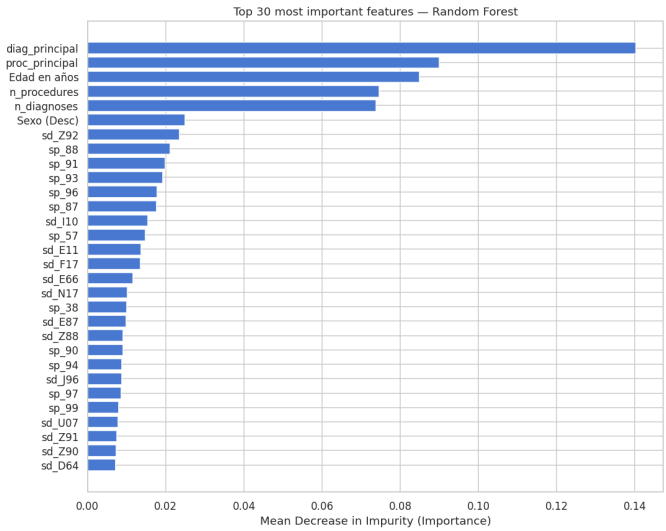

Fig. 4. Top-30 feature importances (mean decrease in impurity) for the Random Forest baseline. The principal diagnosis code dominates, followed by the principal procedure, age, and case-complexity counts.

with the single held-out split within one standard deviation, suggesting that the reported performance is not an artefact of the particular test partition.

The gap between weighted and macro-averaged F1 is notable: on the same test set the macro F1 is only 0.38 (macro precision 0.39, macro recall 0.42). This discrepancy reflects the long-tailed class distribution: frequent GRDs drive the weighted average upward while the hundreds of low-support classes pull the macro average down. Addressing this gap is a priority for the final model.

B. Feature Importance

Figure 4 shows the top-30 features ranked by mean-decrease-in-impurity for the Random Forest baseline. The principal diagnosis code (`diag_principal`) is the single most informative feature with an importance of ≈ 0.14, followed by the principal procedure prefix (`proc_principal`, ≈ 0.09), patient age (≈ 0.085), and the two case-complexity counts (`n_procedures` and `n_diagnoses`, both ≈ 0.075). Together, these five features account for roughly half of the total importance mass.

This ranking is consistent with the clinical definition of the GRD system, in which the primary diagnosis and the presence of a significant operating-room procedure are the main determinants of the assigned group, with age and comorbidity count modulating the severity sub-classification (W/MCC, W/CC, W/O CC). Sex contributes a much smaller but non-negligible share (≈ 0.025), which is expected given its relevance for obstetric GRDs (146xxx, 131xxx).

Among the multi-hot secondary features, the highest-ranked diagnosis indicators correspond to `sd_Z92` (personal history of medical treatment), `sd_I10` (essential hypertension) and `sd_E11` (type-2 diabetes), reflecting highly prevalent comorbidities in the hospital's adult population. The top secondary procedure indicators (`sp_88`, `sp_91`, `sp_93`, `sp_96`) correspond to diagnostic imaging, microbiology, physical therapy, and respiratory therapy respectively — consistent with a secondary-care acute hospital profile.

C. Per-Class Analysis

Per-class F1-scores vary widely. Frequent GRDs with well-defined clinical profiles achieve high performance: the top-performing classes include *146101 — Cesárea* (F1 = 0.80, 163 test samples), *146121 — Parto vaginal con procedimiento* (F1 = 0.78, 128 samples), *044183 — Bronquiolitis y asma W/MCC* (F1 = 0.87), *131111 — Procedimientos sobre útero y anexos* (F1 = 0.85), and *041013 — ECMO ventilación mecánica prolongada* (F1 = 0.86). Conversely, a large number of low-support classes (fewer than ten test samples) obtain F1 = 0.00 because the model never predicts them correctly in the held-out set.

Figure 5 shows the confusion matrix restricted to the fifteen most frequent GRD classes. Two dominant error patterns emerge. First, systematic confusion occurs between severity variants of the same base GRD: 22 of 153 true cesarean deliveries (146101) are misclassified as *146102 — Cesárea W/CC*, and 26 of 44 true cesarean-W/CC cases are in turn predicted as plain cesarean, showing that the baseline model struggles to separate severity sub-groups defined by comorbidity complexity. Similar patterns appear among the vaginal-delivery family (146121 / 146131 / 146122). Second, a smaller number of errors occur across related medical groups with overlapping presentations, such as *044163 — Neumonía simple* being predicted as *041023 — Ventilación mecánica prolongada* in respiratory cases requiring intubation.

These patterns are clinically coherent: severity modifiers (W/MCC, W/CC, W/O CC) depend on the full comorbidity profile, which is only partially captured by our top-50 multi-hot encoding. Richer representations of the secondary diagnoses — in particular, indicators for CC/MCC-defining codes per the FONASA grouper — are expected to close a significant part of this gap.

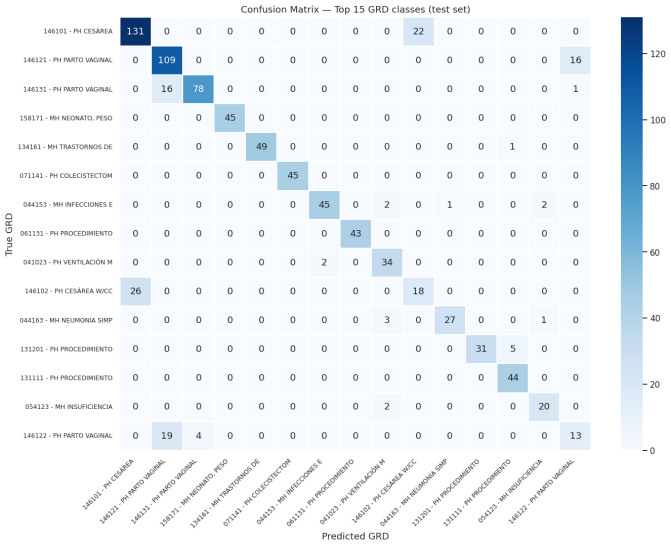

Fig. 5. Confusion matrix for the 15 most frequent GRD classes in the test set. Diagonal entries show correct predictions; the dominant off-diagonal entries correspond to confusions between severity sub-groups of the same base GRD (e.g., 146101 ↔ 146102 cesarean variants).

V. Discussion

The preliminary results confirm the feasibility of ML-based GRD prediction from routinely collected hospital data. The principal diagnosis code is the dominant signal, as expected from the clinical definition of the GRD system. However, secondary diagnoses and procedures provide complementary information, particularly for differentiating severity sub-groups (e.g., W/MCC vs. W/CC vs. W/O CC variants of the same GRD).

The main challenge identified is the *class imbalance*: hundreds of GRD classes appear fewer than ten times in the dataset, making it impossible to learn reliable decision boundaries without additional data or oversampling techniques. The 5-fold cross-validation provides more stable estimates of generalization than a single train-test split for these low-frequency classes.

Future experiments will include: (i) gradient-boosting models (XGBoost, LightGBM) which typically outperform Random Forests on tabular data; (ii) hierarchical embeddings of ICD codes to leverage clinical ontology structure; and (iii) SMOTE-based oversampling for minority classes.

VI. Conclusions

This progress report presents an exploratory analysis and a baseline Random Forest model for GRD prediction using clinical data from Hospital El Pino. The main findings are: (1) the dataset presents high-cardinality, heavily imbalanced GRD classes; (2) secondary columns have increasing but expected missingness; (3) the principal diagnosis category is the strongest predictor of GRD, followed by age; and (4) the Random Forest baseline substantially outperforms the majority-class dummy, demonstrating the viability of the ML approach. Final model selection and comparison will be reported in the complete Phase 1 report.

Limitations and Future Work

The current feature representation uses only the three-character ICD-10 prefix, discarding finer code-level information. Future work will incorporate the full ICD code, text descriptions (via TF-IDF or clinical embeddings), and interaction features between diagnoses and procedures. Extending the study to temporal data (readmission patterns) and multi-site validation would further strengthen the clinical applicability of the model.

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